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Paleomagnetic results from Saudi Arabia and the Permo–Triassic Pangea configuration

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Abstract

The Pangea configuration for the interval of Late Carboniferous to Early Jurassic is still debated: was it similar to the well known Wegenerian Pangea or did Gondwanaland move with respect to Laurussia along a major right-lateral lineament of several thousand kilometres during the Triassic, as suggested by Irving? We try here to answer this important question with the paleomagnetic study of well dated Permo–Triassic sediments from a borehole in Saudi Arabia. Sediments of the same age have also been sampled at the surface, but proved to be remagnetized, probably by a strong desert alteration. The paleomagnetic study of the bore-hole core yielded a high temperature component magnetization thought to be of primary origin carried either by magnetite (Late Permian limestones) or hematite (Early Triassic red shales) with antipodal normal and reversed inclinations. Its co-latitude and pole directions compare well with those from a selection of South American, African, Madagascar, and Moroccan poles. A new reliable pole for west Gondwana with a mean age of 244 ± 11 m.y. is derived ($\lambda = 53.4°$N, $\phi = 259.4°$E, $A_{95} = 3.6$ in West African coordinates), and its comparison with the Laurussian poles of same age window strongly suggests a Pangea in which the northern part of South America was facing the east of North America (Pangea B of Irving) at the Permo–Triassic boundary. A large right-lateral strike-slip movement of some 3500 km during the Triassic is thus required to reconstruct Pangea to its Jurassic pre-Atlantic opening position.

Keywords: paleomagnetism; Pangea; Permian; Triassic; Tethys

1. Introduction

It is often believed that a long period of relative plate motion quiescence followed the Variscan orogeny, after the assemblage of the Pangean megacontinent. This situation may have lasted for a very long period of time of some 100 m.y., from the end of Carboniferous to Early Jurassic. The break-up of Pangea then progressively led to the present distribution of continents and oceans. Paleomagnetic comparisons of Late Paleozoic to early Mesozoic apparent polar wandering paths (APWPs) for Laurussia and Gondwana, however, led Irving [1] to propose an important southwest motion of Gondwana with respect to Laurussia, along a major right-lateral strike-slip zone of some 3500 km with an NE–SW azimuth roughly parallel to the Appalachian mountain range.

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The initial Late Paleozoic configuration was called Pangea B or C, depending how far to the East Gondwanaland was displaced away with respect to Europe [1–3]. The Triassic timing of this motion was at variance with the age of activity of the major right-lateral wrench fault in eastern North America, erroneously dated at that time as pre-Permian by continental sediments. Most reconstructions taking into account this geological data thus favoured a quasi-motionless Pangea, which configuration was similar to that preceding the opening of the Central Atlantic [4,5]. This Pangea was called Pangea A and its configuration was close to the Wegenerian Pangea. The poor quality of Permo–Triassic Gondwanian paleomagnetic data could not overcome at that time the force of geological evidence, and led to numerous debates in the paleomagnetic community itself [6–10]. Several types of intermediate Pangea models (Pangea A2), in which the strike-slip motion between Laurussia and Gondwana was minimized during Triassic were proposed: Gondwana would have undergone only a minor counter-clockwise rotation corresponding to a fit of both Carboniferous and Early Jurassic paleomagnetic poles only, the poles of Permo–Triassic age being redated or even completely excluded [6,7,9,11].

Progress during the last few decades in the stratigraphy of eastern North America [12] have led to a new date for the continental basins of Carnian (Late Triassic); this is supported by both new paleontological and geochronologic data. Ricou [5] then proposed that right-lateral displacement of Gondwana occurred in the Permian and also during the Triassic. However, the amount of Triassic motion he proposed remained small when compared to that indicated by [1] and [3]. Clearly, a definitive answer will not be found until new, well dated, reliable paleomagnetic data from Gondwanaland improve the database. We recall that only a single pole has been published since the end of the 1970s [13]. With this idea in mind, we have carried out a paleomagnetic survey of Late Permian and Early Triassic age sequences from

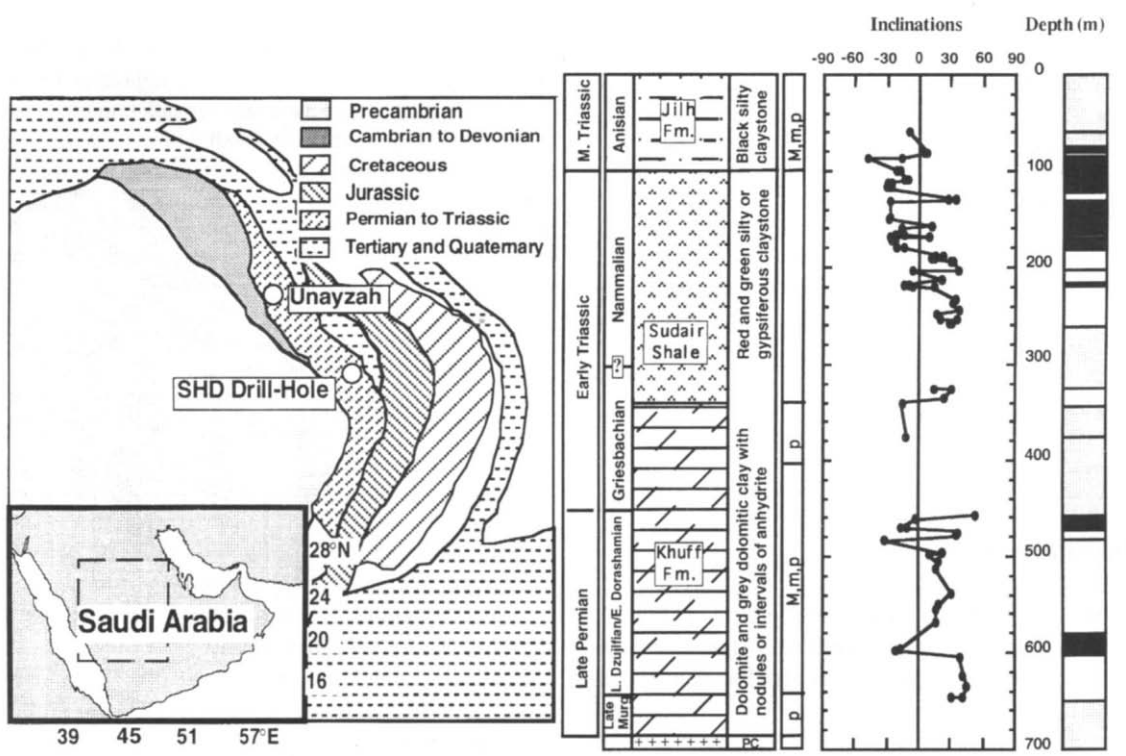


Fig. 1. Simplified geological map of Saudi Arabia showing the emplacement of the SHD bore-hole, and simplified lithostratigraphic log of the SHD bore-hole, magnetic inclinations versus depth, and magnetic polarities: black = normal, white = reversed, shaded = no samples. *M* = macrofauna; *m* = microfauna and *p* = palynology refer to the paleontological dating.

Saudi Arabia, both at outcrop and in a bore-hole. This target was chosen because of the accurate Permo–Triassic dating of the series [14,15], which may greatly help in solving the controversies surrounding the real age of west Gondwanian poles. The geographic situation at the extreme northeast of Gondwana is also important for the improved accuracy of further reconstructions and verification of the geocentric dipole hypothesis at that time.

2. Geological setting of Central Arabia

The Late Permian–Late Triassic succession of central Saudi Arabia is described as the Buraydah Group [14,15]. It is subdivided into four formations, from base to top (Fig. 1):

1. The Late Permian to Early Triassic Khuff Formation, mainly represented by carbonate, evaporite and clay deposits of transitional to marine paleoenvironments. The Formation is dated by floras, microfloras, benthic foraminifers and other marine faunas as Late Permian (Murghabien–Djulfian) and, in its topmost part, to Early Triassic (Scythian).
2. The Early Triassic Sudair Shale is essentially composed of marine evaporitic claystone assigned to a Scythian age, according to its microflora content.
3. The Middle to Late Triassic Jilh Formation, which consists of interbedded sandstone, claystone, and dolomite of transitional to marine paleoenvironments. The formation is poorly dated at the base (?Anisian) and dated as Ladinian to Carnian in its middle and upper parts on the basis of its conodont content.

Samples have been collected from outcrops of the Khuff Formation and Sudair Shale in the Unayzah area in central Saudi Arabia (Fig. 1). We have also sampled the SHD1 bore-hole, fully cored trough Early Triassic and Late Permian deposits [16]. The lithologic log [14,17] is given in Fig. 1.

3. Paleomagnetic results

Natural remanent magnetizations (NRM) were measured using a CTF cryogenic magnetometer in the shielded room at the Institut de Physique du Globe de Paris (IPGP). Stepwise thermal and/or alternating field (AF) demagnetizations were carried out up to 690°C or 100 mT with a laboratory built oven or a Schonstedt GDS single-axis demagnetizer. A mixed thermal/AF demagnetization was sometimes used for some rocks containing magnetite as a remanence carrier. The bulk magnetic susceptibility of the specimens were measured at each demagnetization step in order to monitor magnetic mineral transformations during progressive heating.

3.1. Permian outcrops

Standard cores were taken using a portable gasoline drill and oriented using a magnetic compass. The stepwise Isothermal Remanent Magnetization (IRM) acquisition indicates that saturation is still not reached in a field of 1 T and suggests the presence of high coercivity minerals. The thermal demagnetization of three IRM of different forces (1.26, 0.4 and 0.12 T) applied successively to each of three perpendicular axes of the sample is used to characterize the magnetic carriers [18]: a strong decrease in the magnetization intensity after 120°C (the unblocking temperature of goethite) is evidenced; the unblocking temperature of some 580°C on the soft component indicates a mineral of the titanomagnetite family. The NRM intensity after the first step of thermal demagnetization at 120°C appears frequently to be very low and close to the noise level of the CTF 3-axis SQUID magnetometer. A component of normal or reversed polarity which passes through the origin is sometimes isolated in the 200–430°C temperature range. The mean direction calculated using Fisher statistics [19] appears to be indistinguishable from that of the present centred axial dipole (I_g = 41.5°, D_g = 357.1°, a_{95} = 3.1°, n = 20, compared to D = 0° and I_g = 41.7°) and is suspected to be a remagnetization linked to the intense alteration in the desert environment. Moreover, evidence for water circulation is indicated. Such a mean direction translates to a paleolatitude of some 24°N during the Late Permian and is incompatible with the major trends of paleogeographic reconstruction for that time.

3.2. The SHD1 Permo–Triassic bore-hole (λ = 24°N, ϕ = 45°E)

The bore-hole we have studied encompasses the Khuff, Sudair Shale and Jilh formations (Late Per-

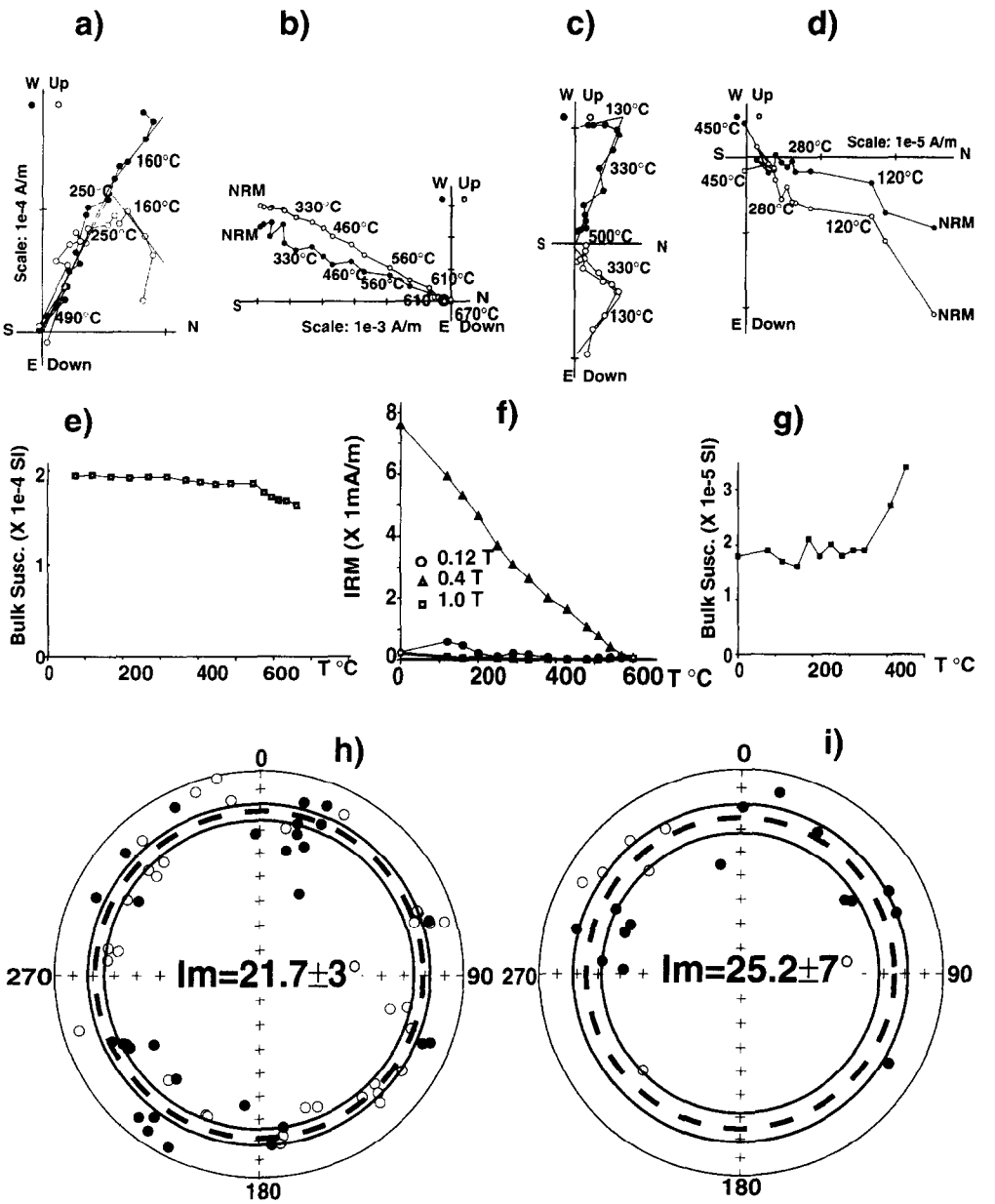


Fig. 2. Orthogonal plots of thermal demagnetization for the Sudair Shale, (a) and (b), and the Khuff Formation samples, (c) and (d). The PEF (when determined) and the characteristic component of magnetisation are indicated by a line. (e) and (g) Bulk susceptibility evolution with increasing temperature of typical Sudair Shale (e) and Khuff Formation samples (g). (f) Thermal demagnetization of three IRM of different forces (1.26, 0.4 and 0.12 T) applied successively to each of three perpendicular axes of the sample [18] is used to better characterize the magnetic carriers. (h) and (i) Equal area projection of the directions derived from the characteristic component from the Sudair Shale (h) and Khuff Formation (i). Closed symbols = downward direction of magnetic vector; open symbols = upward direction of magnetic vector. The mean inclination and associated 95% confidence interval were computed using [20].

mian to Middle Triassic age). Core segments of about 1 m long were recovered with a vertical orientation only. The multiple fractures all along the core did not allow an azimuthal orientation (even relative between nearby segments). A special sampling equipment was developed in the St. Maur laboratory; a V-shaped gutter is used to maintain the core while a frame carrying an electric drill is mounted and allows a precise perpendicular sampling. Sometimes the argillaceous content of the rock was too high to allow water drilling. Small cubes were then directly cut on the core segment itself using a circular saw. The bore-hole was drilled vertical within 1°, while the general dip of the series is small, being a maximum of only 0.7° towards the northeast.

3.2.1. The Sudair Shale (Early Triassic)

This formation consists of red and green claystone. Green claystones were found to be magnetically unstable and are rejected from further consideration. We have sampled 60 specimens of red claystone evenly spaced throughout the 200 m thickness of this facies. In order to try and isolate a present earth field (PEF) component, narrow steps were chosen at the beginning of thermal demagnetization. Orthogonal projections of the magnetic vector during thermal treatment are shown in Fig. 2a,b. One or two components of magnetization could be separated. A first low temperature (LT) one, shown in Fig. 2a is isolated between 20° and 130°C and found in 8 samples only. This component is marked by a clear change in the inclination value, while the change in the declination remains moderate most of the time. Using the McFadden and Reid statistics [20], the mean LT component inclination for these 8 samples is found to be $38° \pm 5.8$. This value is situated between the 41.7° of the axial dipole field and the 35° computed after the 1994 IGRF. A second component is isolated between 200°C and 610°C (and even 680°C for some samples, determining a segment passing through the origin). The magnetic carrier is most probably a mineral of the hematite family, as shown by the high values of the unblocking temperatures Fig. 2a,b. No significant changes in the bulk susceptibility are observed during heating, suggesting that any mineralogical changes that may have occurred were only minor (Fig. 2e). The mean direction is taken in the 200–610°C temperature range. Half of the samples carry a positive inclination (Fig. 2h). The separate means of negative and positive inclinations ($I_m = 22.1° \pm 5$ for the positive inclinations and $I_m = 21.3° \pm 3.6$ for the negative inclinations) are antipodal (they only differ by 0.8°). A mean inclination of $22° \pm 3$ is derived for the 60 samples, using the statistic method of [20]. No particular trend with respect to time can be evidenced.

3.2.2. The Khuff Formation (Late Permian–Early Triassic)

The late Djulfian and early Dorashamian parts are represented by carbonates and clay deposits of the Khuff Formation. In total, 38 samples were sampled in the limestones. One or two components of magnetization could be evidenced (Fig. 2c,d): a first one in the range 20–160° gave a mean direction of $40.1° \pm 4.3$ for $N = 8$ samples. This direction is again close to the PEF directions. Another component was often isolated in the 160–580°C temperature range and passes through the origin (Fig. 2c,d). The thermal demagnetization of the triaxial differential IRM presented in Fig. 2f shows that the medium coercivity component (which carries all the intensity) falls at 580°C. This suggests the presence of a mineral of the titanomagnetite family. The direction become often erratic after 370°C and can be attributed to two factors: first, the initial intensity of some of these samples appears to be rather low (from 2×10^{-6} Am^{-1} to 4×10^{-5} Am^{-1}) and close to the noise level of our cryogenic magnetometer (5×10^{-6} Am^{-1}), preventing us from obtaining thermal demagnetization diagrams until 580°C. Second, a mineralogical transformation is marked by an increase in the susceptibility after 350°C (Fig. 2g). HT directions were isolated on 21 samples. Six of the inclinations are of an opposite sign to the remaining 15 (Fig. 2i). The mean paleoinclination, using the statistics of [20] is $25.2° \pm 7$, and is indistinguishable from that found in the Sudair red shales at the 95% probability level.

Since we have no azimuthal orientation of the core, the mean inclination may represent a North or South paleolatitude of the sampling site. The hemispheric ambiguity can be solved in our case by using the declination of the LT component (relative to present geographic north) to rotate the declination of

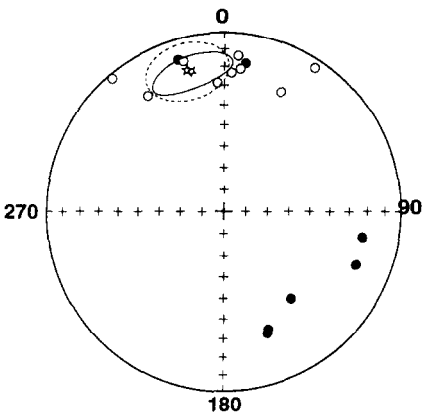


Fig. 3. Characteristic directions of the 16 samples containing a PEF component and rotated using this component. (Equal area projection).

the HT component for each sample. No tilt correction is necessary since the general dip of the series is nearly 0°. Sixteen samples were shown to display a LT component. The Fisher mean of the rotated values for the 16 corresponding HT component directions is D = 165°E, I = 19°N, a = 15.9, k = 6.4 (Fig. 3). This rather large confidence interval is due to a large dispersion in the declinations. This is illustrated by the use of the bivariate statistic [21], showing that the inclinations are well grouped: D = 167°, I = 20°, ax = 9, kx = 15, ay = 17, ky = 4. Since the autochthony of the Central Saoudian Series is beyond any doubt, the deposition in the southern hemisphere is certain.

The Late Permian part of the section is determined by a small number of samples (21) which show a predominantly reversed polarity, while the Early Triassic sequence, based on the inclinations of 60 samples, shows 15 magnetic intervals. Upper Permian stages are not yet perfectly correlated throughout the world. Problems of intercorrelation are, nevertheless, very important for further discussion and we decided in this paper to use the time scale of Harland [22], who use indifferently, eastern Europe, Tethyan and southwest North American stages. This time scale is also used by McElhinny and Lock [23] in their paleomagnetic database widely used in the next chapter. The late Dzhulfian–early Dorashamian age of our section is older than the Capitanian and probably contains the lower part of the Changsingian [24]. We have used the same stages as [22] for the Triassic.

3.3. A Permo–Triassic boundary pole for western Gondwana

The small number both in time and space of reliable pole positions for each of the main blocks constituting the Gondwana continent does not allow the construction of a well constrained APWP for each of them. It is therefore natural to turn to the best data of the same age observed on each of the blocks and to transfer them onto one plate by using oceanic kinematics parameters. Reliable models of ocean closure are available and can be used safely, at least for the resolution that is required. An additional difficulty is that plates cannot be considered as rigid throughout geological time. Africa and South America have been shown to have been internally deformed when they separated by several authors [11,25–29]. A paleomagnetic demonstration of the Cretaceous deformation in Africa has been made by [30]. These authors showed that a good agreement between poles was obtained by using the set of rotation parameters of [11] to transfer data from South America, South Africa, West Africa, Northeast Africa, Madagascar and Arabia (Table 1). This was also discussed by [29], who underlined that these parameters probably represented a minimum estimation of the intra-plate deformation at that time.

Due to the small number of studies, the rejection criteria we have used are not very stringent: we have

Table 1
Parameters used in transferring paleomagnetic poles

	Latitude	Longitude	Angle	References
Arabia to South Africa	−45.23	−163.72	7.12	[11]
North America to Eurasia	88.5	27.7	38.0	[54]
Madagascar to South Africa	−12.53	−54.96	17.46	[11]
North West Africa to South Africa	9.34	5.7	7.82	[11]
North East Africa to South Africa	−16.3	41.71	2.53	[11]
South America to South Africa	46.82	−30.54	55.88	[11]

retained poles with a 95% confidence interval of less than 17° and with evidence for successful alternating fields and/or thermal demagnetization. We have also attempted to spot evidence of remagnetization, to check the quality of age determination and to assess critically tectonic structure. We have retained studies with more than 15 samples. We have not retained, as [31] did, poles from Adria. The idea that this peninsula was part of stable Africa is far from being proved by paleomagnetic data and, even more, is questioned by the tectonic history of the Mediterranean Tethys [32]. Rather similar pole positions are found during the Early Permian and 'middle' Cretaceous, but important discrepancies are also evidenced during the Late Jurassic that most probably exceed the errors of the ill-defined Late Jurassic segment of the African APWP.

The selected poles for South America, Africa and Madagascar are listed in Table 2 and plotted in Fig. 4a. The pole from the Amana Formation Rio Chasquil in Argentina of [33] corresponds to the analysis of red sandstones in a quasi-monoclinal series. In total, 59 specimens (over 87 samples) have been correctly demagnetized (the pilot study comprises thermal demagnetization between 150°C and 650°C). In situ directions are remote from the PEF direction (I = 12° compared to $I_{PEF} = -47°$), and normal and reversed polarities are found for antipodal directions. Moreover, the directions cluster better after bedding correction. Another study of Permo–Triassic red beds from the Paganzo Formation in Argentina was carried out by [34]. Thermal cleaning led to a better grouping of directions with two polarities, but with some intermediate directions, which may represent uncleaned directions rather than transitional polarities as claimed by the authors. Two pole positions statistically indistinct are computed either for the reversed polarity only or with a 40° VGP cut-off to overcome this problem. Bedding attitudes are fortunately given, and show that directions are significantly different from the present earth field when uncorrected for bedding. The comparison of the di-

Table 2
List of selected poles

Age	λ	ϕ	$\lambda 1$	$\phi 1$	a_{95}	Rock Unit and site # in Fig. 4a	References
South America							
237.5	83.0	137.0	51.0	249.6	8.5	Argentina, Amana Fm. (1)	[34]
243.0	81.0	102.0	55.8	246.2	6.0	Argentina, Mendoza l. South Nihuil (2)	[35]
249.5	77.0	355.0	60.9	279.0	6.0	Argentina, Amana red beds (1)	[35]
250.5	85.0	244.0	48.0	263.8	7.6	Argentina, Amana Rio Chasquil (1)	[33]
250.5	86.0	114.0	53.4	254.1	14.0	Brazil, Corumbatai Fm. (3)	[38]
256.5	82.7	234.7	45.5	263.1	3.4	Brazil, Irati Fm. (4)	[40]
Mean	89.0	110.9	52.9	259.0	7.31	N = 6	
Africa							
235.0	54.0	260.0	54.0	260.0	6.0	Angola, Cassanje series (5)	[47]
238.0	69.8	251.6	62.5	259.0	11.7	Morocco, combined (6)	[49]
238.0	72.0	254.5	64.0	261.5	7.3	Morocco, Ait–Aadel Dolerite (7)	[48]
238.0	77.3	140.6	80.9	178.5	6.5	Morocco, Issaldin Dolerite (7)	[48]
254.0	26.0	267.0	26.0	267.0		Tanzania (8)	[13]
Madagascar							
248.5	65.8	291.6	57.2	259.9	6.1	Sakamena Group (9)	[44]
Poles calculated in this study							
Age	λ	ϕ	a_{95}				
244.0	53.4	259.4	3.6	Mean pole for west Gondwana in South African coordinates			
244.0	60.7	254.3	3.6	Mean pole for west Gondwana in West African coordinates			
244.0	56.7	240.5	16.0	Mean pole calculated from the 16 directions (HT component of the 16 samples showing a LT component) in South African coordinates			

Ages are in million years, λ = latitude (°N), Φ = longitude (°E), a_{95} = error of the VGP; $\lambda 1$ = latitude (°N), $\phi 1$ = longitude (°E) of VGP transferred onto South Africa using the parameters of Lottes and Rowley [11].

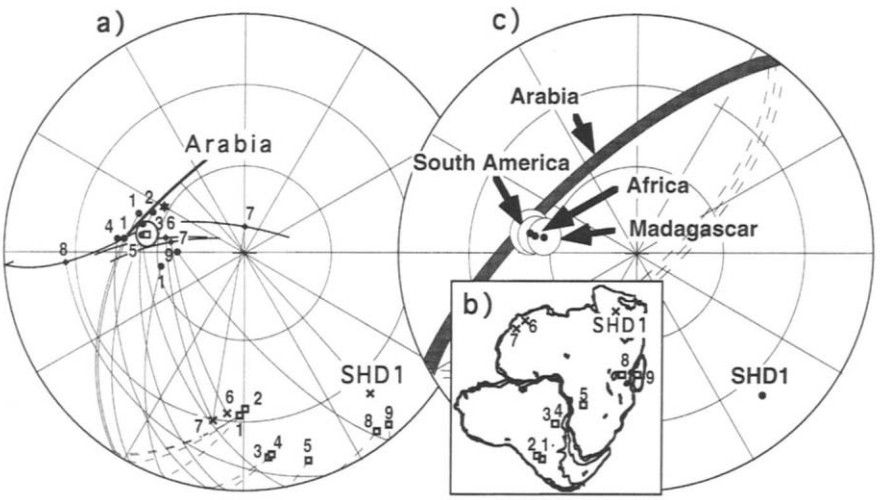


Fig. 4. (a) Mean west Gondwanian paleomagnetic pole (□) and its A95 circle of confidence (shaded area). ● = selected poles of west Gondwana and site locations (□ = Northern hemisphere, x = Southern hemisphere). A small circle joins the sites and their corresponding pole. Heavy line = locus of possible pole positions for allochthonous sites. The black star represents the Arabian pole calculated from the mean of the 16 directions rotated using the direction of the PEF (see text). (b) Location of the sites on a simplified map of west Gondwana. (c) Comparison between autochthonous poles of South America, Madagascar and Africa and the circle of possible pole positions for Arabia.

rections for the Late Permian/Early Triassic show a strong cluster after bedding correction, suggesting a positive fold test not recognized by the authors. Other studies in Argentina have also been performed on red sandstones taken in the Paganzo Formation of Villa Union (22 samples) and South Nihuil lavas in the region of Mendoza (4 sites, 41 samples) [35]. Both studies were thermally and/or AF demagnetized, and yielded normal and mixed polarities, respectively. The age of these formations is not as imprecise as generally believed. Clastic sediments from the Precordillera and Pampean ranges are dated from Late Carboniferous to Middle Triassic [36]. Given that the age of the end of the long reversed Kiaman superchron is not yet accurately known, a range between 264 m.y. and 255 m.y. can be safely assumed [37]. The presence of normal polarity in the above-mentioned studies may thus indicate a post-Kiaman age. Furthermore, stratigraphic correlations by [36] indicate an age younger than the intercalated 232 m.y. lavas found by [38] in the Ischigualasto–Ischichuca basin and Puesto–Viejo Formation. The age of these series is most probably within a maximum range of less than 30 m.y., roughly centred on the Permo–Triassic boundary. The Upper Paleozoic series of the Parana basin (mainly in Brazil) have also yielded paleomagnetic results for the time under consideration [39,40]. Two formations of the Passa Dois Group have been investigated in the same region (Parana basin). The calcareous and marly Irati Formation (54 specimens from 15 hand samples in 6 stratigraphic levels) and the overlying Corumbatai Formation red beds (46 specimens from 16 hand samples) have been dated stratigraphically as no older than Artinskian and from Kazanian to late Tatarian, respectively. Both suffered thermal and alternating field treatments. A single normal polarity in the first section and three reversals in the second may indicate a post-Kiaman age of deposition. Magnetic directions before bedding correction are unfortunately not provided. These Late Permian pole positions coincide well with the Permo–Triassic pole from Argentina. The most relevant problem is in fact to separate recent overprints from the primary component since South America was most probably located for a long period (Late Permian–Cretaceous) close to its present position with respect to the geographic poles [34,41]. In particular, there is the possibility of remagnetization, due to the important Cretaceous volcanic activity prior to the opening of South Atlantic Ocean in the vicinity of the Parana Basin. However, most data older than Late Permian

are of reversed polarity and the age of magnetization acquisition is consistent with the reversed Kiaman superchron. Moreover, a track is observed between poles of Late Carboniferous to Triassic age in the red sandstones of Argentina, rendering remagnetization unlikely Although the South American path is probably not as badly defined as one might have thought, additional paleomagnetic studies are badly needed.

A Permo–Triassic pole of good quality comes from the Sakamena Formation of the Karoo Supergroup of Madagascar [42–44]. This red sandstone continental formation includes marine intercalation, and is thus securely dated simultaneously by brachiopods, ammonites, vertebrates and vegetal paleontology. The paleomagnetic pole (7 sites, 31 samples) was derived from samples that underwent thermal demagnetization with sometimes the addition of alternating fields, and yielded both normal and reversed polarities with antipodal directions. Moreover, the correction for the attitude of the beds resulted in increased precision but, unfortunately, this was not statistically significant because the beds were almost flat-lying.

Six poles from Africa also fulfil our criterions. The Maji Ya Chumvi pole of [45] is dated to the Permo–Triassic boundary by plant fossils, this age being furthermore secured by an overlying marine horizon of Early Triassic age. Thermal analysis revealed a stable component of normal polarity only. No fold test is available because of regional flat bedding. This pole was discussed [46] on the basis of a study in Southern Africa, which suggested an Early Jurassic remagnetization. This clue was based on a negative fold test held by a single site (more than 7 samples) comprising only 3 cores and an $\alpha 95$ of 26°. However, this study cast some doubts about the Maji Ya Chumvi pole and we finally did not retain this study. We have used for Africa the pole of [47] in Angola. This study of rocks of the Karoo system of Angola is dated by fish and plants to the Early Triassic. Nearly antipodal normal and reversed polarities were found by thermal demagnetization of 25 hand samples. Three Middle Triassic poles from Morocco are also available, with results from Issaldin (8 sites, 46 samples) and Ait–Aadel dolerites (15 sites, 62 samples), dated stratigraphically by intercalation of marine levels [48]. Rocks were demagnetized by alternating fields and by thermal demagnetization, and revealed only normal polarities. Various sedimentary and volcanic horizons (5 sites, 54 samples) were also combined by [49] to obtain a pole: the rocks were demagnetized either by AF or thermal treatment of a rather good quality, although all their samples are also of normal polarity. Finally, the pole from [13] is the most recently published (1993), and provides a sufficient number of sites plus a positive fold test. It furthermore includes older data from the same region [50].

All our selected poles were transferred onto South Africa and are shown on a stereographic projection (Fig. 4a). We have also plotted the locality of the regions of sampling (open squares). Most of the selected poles appear to be well grouped in a region situated close to a 260°E longitude and a latitude of 55°N. Clear outliers in Fig. 4a are the poles of Tanzania [13] and Morocco [46]. In both cases they can be reconciled with the bulk of the data if one rotates them along a vertical axis: this is illustrated on Fig. 4a, which shows small circles centred on their respective sites of sampling passing through the middle of the cluster. These local rotations in both regions may be linked to later tectonics. In the case of Tanzania [13], sampling sites may have been rotated during Cenozoic rifting. A close look at the sampling sites on a geological map of this area shows that most sites stand in a narrow band of strike-slip faulting which follows Lake Tanganyika and ends near Lake Malawi. It is obvious that these sites may have been significantly rotated. We remark that the VGP of the Mikumi site, the most remote from this deformation area, lies the closest to our Gondwanian poles. The Morocco regions north of the Tizin strike-slip faults have also been affected by large scale strike-slip faulting several times since the Permian and even recently [51]. Again, a simple rotation along a vertical axis helped to reconcile these data with the Permo–Triassic cluster. South American data from Argentina may have also suffered slight clockwise rotations as a result of recent deformation linked to the active tectonism of the Andean cordillera in the last 20 m.y. (Roperch and Chauvin, pers. commun., see also [52]). The position of the South American data in Fig. 4a is nevertheless close to the bulk of data.

We may now compare these data with the well dated directions of Arabia, which represent a signifi-

cantly different part of Gondwana. The mean pole computed after rotation of the present earth field declination (black star in Fig. 4a) is shown to be in very good agreement with other poles. However, we do not use it for computing the mean P/T pole of west Gondwana. The errors involved are a cumulation of those of the determination of the present earth field, its unknown distribution around the present geographic pole and the classical errors in the HT component direction. Moreover, the low number of samples (16), and the large A95 (16°) are only marginally acceptable by our own criterion selection. We have used the mean paleocolatitude small circle based on 60 samples, and computed using the statistic of [20].

Fig. 4a shows an excellent agreement between data of Permo–Triassic age from western Gondwana. Our mean P/T pole for western Gondwana is computed using a combination of VGPs and small circles, using McFadden and McElhinny's method [53]. Small circles have been used for mobile zones (Morocco and Tanzania) and Arabia, while VGPs are used for other sites: we obtain a pole with a mean age of 244 m.y. lying at $\lambda = 53.4°N$, $\phi = 259.4°E$, $A_{95} = 3.6$, $k = 138.0$ in South African coordinates and $\lambda = 60.7°N$, $\phi = 254.3°E$ in West African coordinates. The 95% confidence interval, on the order of 4°, shows the very good coherency of the originally selected paleomagnetic data, and plate kinematic models compatible to that level of accuracy. Moreover, this gives an idea of the good quality of the geocentric dipole assumption for this period. We also show in Fig. 4c the distribution of mean autochthonous VGPs from South America Madagascar Africa and the Arabian small circle, which show an excellent cluster of the data. This compilation, thus, does not lend support to an age error in the Permo–Triassic poles for western Gondwana as claimed by some authors. The arithmetic mean of individual age errors is 11 m.y. This last result is a quite important clue for solving the problem of the Pangea reconstruction at the Permo–Triassic boundary, as discussed below.

4. Discussion

A high resolution Paleozoic and Mesozoic APWP for Laurussia has been recently established by Van der Voo [9]: reliable poles for Europe (and Russia west of Urals) have been transferred onto North America using available rotation parameters, and then averaged with North American poles of corresponding age. Van der Voo demonstrated that the best fit is that of Bullard et al. [54]. This fit ensures the best general agreement between the two APWPs from the Cambrian to Jurassic. New paleomagnetic data both in Europe and North America [55–57], which have been published since the work of Van der Voo, led us to recompile the Laurussian APWP (Table 3). We have used the 190–310 m.y. poles of western Europe and North America available in the global database of [23]. Contrary to Van der Voo, we do not use poles from Russia, for which there is not enough detailed information. Criteria more stringent than those applied for west Gondwana were applied: sites must lie outside tectonic zones, no paleomagnetic test (fold, reversal or conglomerate) must be negative, the error in age must be smaller than 15 m.y., the confidence limit must be smaller than 15° and the number of samples greater than 20. Poles from western Europe were transferred onto North America using the fit of Bullard et al. [54] and poles from the Colorado Plateau were rotated using the Euler pole of [58], as discussed in [9]. The data have been sampled in 14 m.y. non-overlapping windows. This width was chosen as the minimum presently permitted by the number and quality of the available data. This APWP is compared to that of Van der Voo [9] in North American coordinates on Fig. 5a. They both follow the same track between 280 and

Table 3
APWP for Laurussia between 199 and 297 m.y. in North American coordinates

Age (m.y.)	λ	ϕ	a_{95}	N
199.0	64.5	94.0	3.0	13
214.0	60.8	96.3	9.7	6
230.0	53.9	101.4	5.2	6
242.0	51.4	111.3	3.4	19
255.0	46.8	124.0	4.2	14
267.0	44.0	126.7	5.5	6
284.0	41.1	125.5	4.5	9
297.0	40.8	126.6	5.3	8

λ = latitude (°N); Φ = longitude (°E); a_{95} = error of the VGP; N = number of data used.

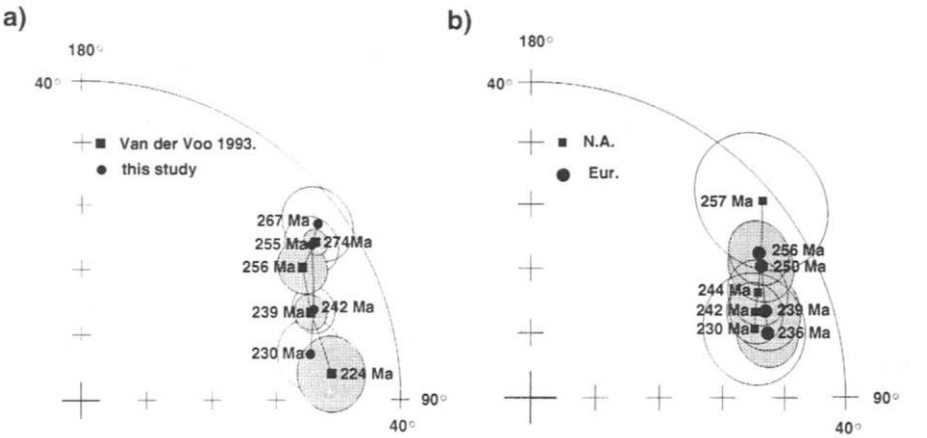


Fig. 5. (a) Comparison of our APWP for Laurussia and the one of [9]. (b) Comparison of the APWP of Western Europe and North America for the same period. (All poles in North American coordinates).

220 m.y. and a given age mean pole for one track is contained within the 95% error bar of the other. Two APWPs of Europe and North America using a 20 m.y. window are also shown in Fig. 5b. The respective means for a given age are contained in the respective 95% ellipses of confidence. Only, the 257 m.y. pole from North America is ill defined but the large radius of its ellipse of confidence contains the mean 256 m.y. pole of Eurasia. We thus do not detect a significant difference in pole positions between North America and Europe with respect to age, as suggested by [7], in the time window considered. The mean poles of Fig. 5b shows that an important drift occurred between 256 and 230 m.y. (the angular distance between these two poles is 16°). We recall that our Gondwana pole is dated at 244 ± 11 m.y. (255–233 m.y.) This time period is critical because a small error in dating leads to a significant error in Laurussian pole position. We thus compare our west Gondwana pole and the APWP segment of Laurussia between 256 (± 10) and 228 (± 9) m.y. Fig. 6a shows our mean west Gondwanian paleopole and the APWP of Europe relative to a North American fixed Pangea A or A2 reconstruction, using the most frequently used parameters between North America and West Africa. In both cases a large discrepancy between the pole of each APWP is obvious and an important relative motion between Gondwanaland and Laurussia is required. Fig. 6b shows two possible reconstructions where the west Gondwanian paleomagnetic pole at 244 ± 11 m.y. is fitted on the Laurussian poles of 230 ± 9 and 242 ± 5, while Fig. 6c shows a fit with the 256 ± 10 m.y. Laurussian poles. All these comparisons are valid within the age uncertainties of our Gondwanian pole. A configuration similar to that of Pangea C [2] is obtained by fitting our Gondwanian pole with the 256 ± 10 m.y. pole of Laurussia (Fig. 6c). A large ocean remains opened in the eastern part of North America during the Permo–Triassic. This scenario, which does not take into account at all the Permo–Carboniferous history in the Appalachian region, appears very unlikely.

In Fig. 6b we have maintained a Gondwanian–Laurussian contact in the Appalachian region. Both reconstructions show that the western coast of South America is facing the eastern side of North America and recall the Pangea B of Irving [1]. The two Gondwana–Laurussia finite Eulerian poles of these reconstructions are at $\lambda = 38.09°S$, $\phi = -177.49°E$, angle = 69.90° and $\lambda = -37.04$, $\phi = -173.12°E$, angle = 67.53°, respectively, giving similar respective motions of 3700 and 3500 km on the strike-slip fault boundary between both land masses to join their well known Early Jurassic relative positions before the opening of the Central Atlantic. The Mauritanides are situated south of present Black and Caspian seas, and the Tethyan region is completely closed. This solution is compatible with the collisional character of the Appalachians, Mauritanides

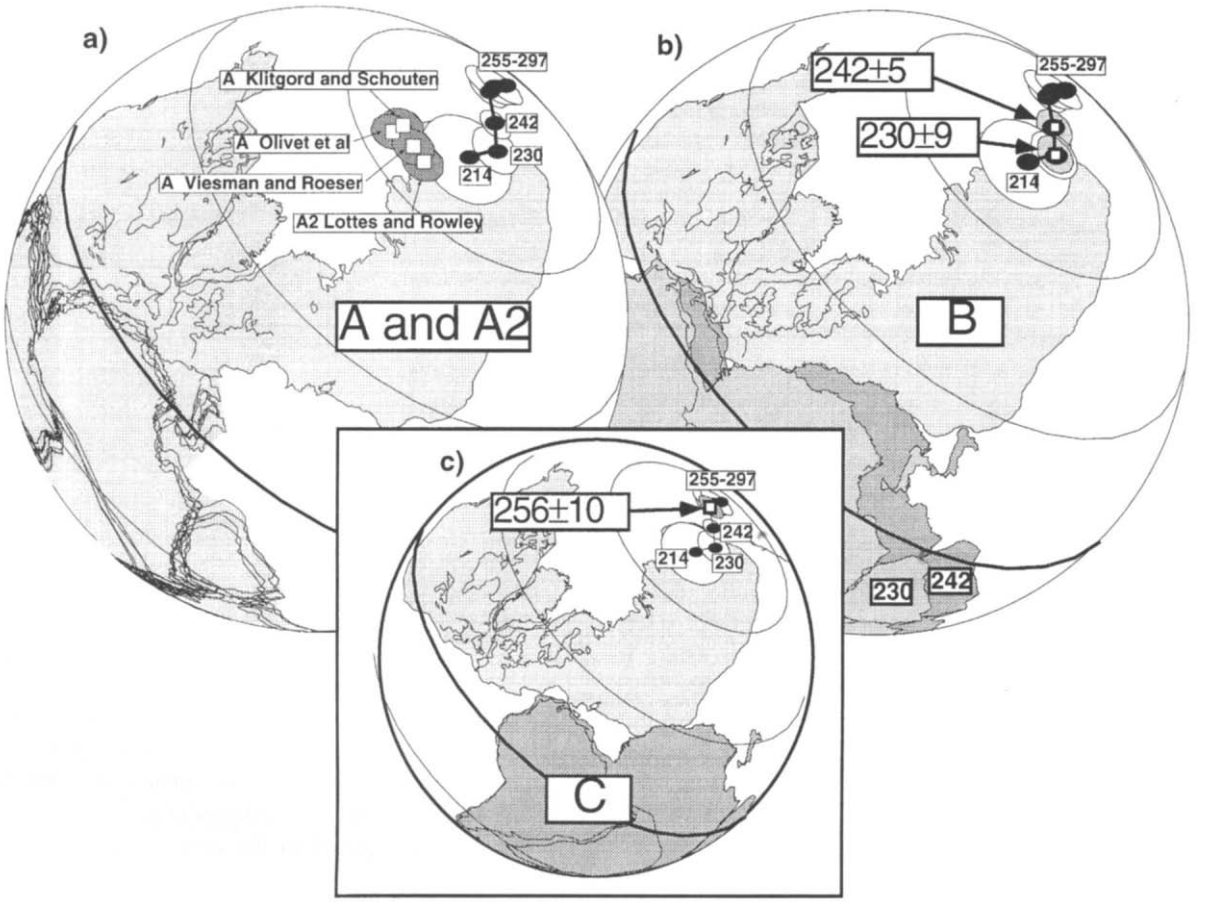


Fig. 6. (a) Comparison of our 244 ± 11 m.y. west Gondwana pole with poles of Eurasia. A Pangea of type A or A2 (see text) is reconstructed using several Central Atlantic fit parameters [61–63]. The large observed discrepancy between Laurussian and Gondwanian poles requires an important relative motion between the two land masses. (b) Fit using our 244 ± 11 pole of western Gondwana and the 230 ± 10 (Gondwana in light grey) and 242 ± 10 m.y. poles of the Laurussian APWP (Gondwana in dark grey). Both are Pangea of B type. (c) A Pangea of Type C, required by the fit of our pole and the Laurussian one dated at 255 ± 10, if one wants to avoid unacceptable overlaps. A large ocean remains east of North America. This situation is not supported at all by geologic data.

and Variscides [4]. The Triassic timing of the lateral motion is compatible with the general extensional pattern in the Tethyan region, marked in particular by the Ladinian (Middle Triassic) age of the basins in the Mediterranean area [5,32]. A precise quantification of the velocity along-strike on the fault is not a simple matter since it would require a Late Permian to Early Jurassic paleomagnetic analysis with a high level of detail. A rough estimation can, however, be attempted: the end of the motion is possibly indicated by the good agreement between Late Triassic–Early Jurassic poles of Africa, South America and Europe [59]. Taking a maximum displacement of 3700 km and a duration of 40 m.y. corresponding to the approximate duration of Middle and Late Triassic would require a velocity of some 9 cm/yr. Even doubling this value (taking for example a 20 m.y. duration only) would still be comparable with the well known 17 cm/yr northward motion of India before its collision with Asia [60].

5. Conclusion

Our study from Arabia, along with a selection of older published data for west Gondwana, yield a

robust paleomagnetic pole of Late Permian/Early Triassic age (244 ± 11 m.y.). The cluster of paleomagnetic west Gondwanian directions derived from a large geographic distribution of sites indicates that the assumption of a geocentric dipole field is valid for this period. The same conclusion can be drawn from the APWP of Laurussia, very well constrained during this time segment. The comparison of our pole from Gondwana with the corresponding time segment of the Laurussian APWP indicates that a Permo–Triassic Pangea of type B is the most likely solution. A Triassic 3700 km motion of Gondwana with respect to Laurussia is thus required to reach the Early Jurassic Wegenerian Pangea.

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